



ET 431 Circuit Analysis I, 4 Credits

UNH Credit Hour Policy:

This syllabus reflects the federal definition of a credit hour, which entails a minimum 3 hours of engaged time per week per credit over a 15-week semester. Examples of engaged time include class time, assignments, examinations, laboratories, participation in course-related experiences (attending a talk or performance, speakers and events, fieldwork, etc.), conferences, and office hours. Student work reflects intended learning outcomes and is verified through evidence of student achievement. For more information, please see: [NECHE policy on Credits and Degrees](#).

Attributes: NONE

CLASS MEETING INFO: (fictitious day and time slots)

Section	Day	Start Time	End Time	Room
Lecture	Mon/Wed	10:10 AM	11:30 AM	P102
Lab	Monday	11:40 AM	1:00 PM	P114

Course Description:

First course in electronic circuit analysis exploring the fundamental idea of current and voltage. Topics include the basic laws and theorems that govern simple electrical systems, Kirchhoff's laws, Ohm's law, power relationships, resistance, inductance, and capacitance. Laboratory exercises will introduce the student to the basic measurement techniques of electronic systems using circuit building, power supplies, multimeters and oscilloscopes. This course will also introduce simple circuit simulation techniques.

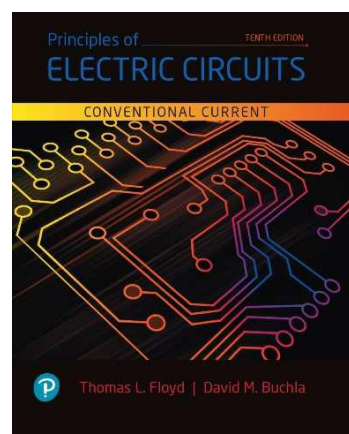
REQUIRED TEXT:

Principles of Electric Circuits (10th Edition)

Thomas L. Floyd

ISBN-13: 978-0-13-487948-2

ISBN-10: 0-13-487948-1



Instructor: Donald Marcotte (Adjunct professor)



e-mail: Donald.Marcotte@unh.edu

Office Hours: Arranged by appointment.

Grading: The grading for this course will be divided among the following categories:

Homework	20%
Tests (2)	30%
Laboratory Exercises	30%
Final Exam	20%

Spreading the overall course grade across different assessment methods is helpful for most students who may have difficulty with one method, e.g. test anxiety, time constraint, etc. It is important that the student complete assignments in each of the above categories in order to achieve a passing grade in this course.

If you have questions during the semester, please feel free to ask the instructor. If you are having trouble understanding the material or with the course, please let the instructor know as early as possible.

Grading Scale:

Percent	Grade	Grade Points	Level of Achievement
94-100	A	4.0	Excellent
90-93	A-	3.67	
87-89	B+	3.33	
84-86	B	3.0	Superior
80-83	B-	2.67	
77-79	C+	2.33	
74-76	C	2.0	Satisfactory, Competent
70-73	C-	1.67	
67-69	D+	1.33	
64-66	D	1.0	Marginal
60-63	D-	0.67	
59-Below	F	0.0	Failure

Assignments: Homework and activity/exercise assignments will be assigned throughout the semester. All assignment information will be posted on Canvas with applicable due dates.



Typically, a two-week advance notice for a test will be given. All assignments must be submitted by the due date. Late assignments will be accepted up to “ONE-WEEK” (5 weekdays) beyond the announced due date. Late assignments will be reduced one full letter grade. Late assignments submitted after the “ONE-WEEK” period will not be accepted and a grade of “zero” will be recorded for that assignment. Tentative dates for tests are indicated on the course outline. Watch for update/revisions to course outline schedule and announcements in Canvas. The schedule will change over the course of the semester.

Assignment Submission Requirements: General requirements for assignments, as well as tests, and final project:

1. Always show all your work. (No partial credit can be given unless steps are clearly shown.)
2. Clearly identify final answers.
3. Work should be legible and performed in a clear, organized manner.
4. Turn in homework in electronic form. Electronic form should consist of a single coherent legible document for each single assignment. Do not submit several pieces of electronic items such as phone snapshots for a given single assignment. Scanning may be done using the ET Commons copier/printer.
5. Include a cover page with assignment information on each assignment submitted. This includes the assignment number and type of assignment, e.g. Homework #1 Chapter 1, Activity 1 – Brainstorming Exercise. Be sure to include your name (and names of your group if applicable) on the cover page.
6. Some assignment formats may vary per activity/exercise.

These items facilitate fair grading of the assignment and granting of partial credit where warranted. Working on homework and activity assignments in groups is permitted but each person will submit individual assignments when required. Activity assignments may be individual or by group depending on the exercise and requested format.

Attendance: Class attendance is important for your learning. You are responsible for all course assignments and meeting all deadlines unless exceptions are agreed upon with the instructor ahead of time. Regular class attendance and participation in discussions is strongly encouraged and expected. If you need to miss class for a planned activity, let me know ahead of time, as early as possible. Students are responsible for all material covered in class and posted on Canvas.

The University absent policy is found in the handbook at this website address:
<https://catalog.unh.edu/undergraduate/academic-policies-procedures/attendance/>

Course Platform: Information on Canvas, the learning management tool we use for this course, may be accessed through mycourses.unh.edu. You will need to access the Canvas site on a regular basis throughout course to access important materials and announcements relevant to the course.



Tests, Lab reports and Final exam: Tests will typically take $\frac{1}{2}$ to 1 hour of class time. (If circumstances dictate, part of a test may be given in a Take Home format.) If a student needs to miss a test for any reason, the instructor should be notified prior to the session in which it is given. In extreme circumstances and if no prior notification is possible, the student must contact the instructor within 24 hours of the missed test and submit a valid reason why the test was missed. A modified test may be given on a case-by-case basis. See the Student with Disabilities section if modified exams are required.

Academic Honesty: In the preparation and presentation of any assigned work – including examinations, tests, quizzes, term papers, reports, papers and other written or oral exercises – every student shall conform to a strict standard of academic honesty. Any attempt to deceive a faculty member or to help another student to do so will be considered a violation of this standard. In all assignments, students must acknowledge the words and/or ideas of others taken from print or electronic media, whether a direct quotation or a paraphrase; any omission of this is dishonest.

Cheating on examinations or tests consists of knowingly giving, receiving, or using – or attempting to give, receive, or use – unauthorized assistance during an examination or test. A faculty member may record a grade of “zero” for any assignment on which a student has plagiarized or cheated. For repeat offenses within a single course, the faculty member may record a grade of “F” for the course. Violations of this policy in multiple courses may result in dismissal from the college. Updated link for Academic Integrity policy for SY 25/26: [Academic Integrity Policy link](#).

Accessibility Services – Manchester Students with Disabilities: The University is committed to providing students with documented disabilities equal access to all university programs and facilities. If you think you have a disability requiring accommodation, you must register with the Student Accessibility Services (SAS) office. Please contact Student Accessibility Services (SAS) located on the Manchester campus in the Student Services Suite (Office 405A).

Accommodation letters are created by our Coordinator with the student. Please follow up with your instructor as soon as possible to ensure timely implementation of the identified accommodations in the letter. Faculty have an obligation to respond once they receive official notice of identified accommodations but are under no obligation to provide retroactive accommodations.

For more information refer to www.unh.edu/sas or contact SAS at 603.862.2607, 711 (Relay NH) or sas.office@unh.edu.

Confidentiality and Mandatory Reporting of Sexual Violence or Harassment:

The University of New Hampshire and its faculty are committed to assuring a safe and



productive educational environment for all students and for the university as a whole. To this end, the university requires faculty members to report to the university's Title IX Coordinator (Bo Zarycky, Bo.Zarycky@unh.edu, 603-862-2930/1527 TTY). Faculty, staff or students on the Manchester campus can also contact Lisa Enright, Deputy Title IX Coordinator (lisa.enright@unh.edu; 603-641-4336; Room 439) to report any incidents of sexual violence and harassment shared by students.

If you wish to speak to a confidential support service provider who does not have this reporting responsibility because their discussions with clients are subject to legal privilege, you can contact the SHARPP Center for Interpersonal Violence Awareness, Prevention, and Advocacy at (603) 862-7233/TTY (800) 735-2964, as well as, Caroline Young, SHARRP Center Advocacy Expanded Services Coordinator for UNH Manchester (caroline.young1@unh.edu; room 417; Available in person Mondays 9 am to 4-pm and available by appointment (in person and virtually) by emailing caroline.young1@unh.edu). Individuals can also access Reach Crisis Services NH 603-668-2299 (24 hours), 77 Sundial Ave., Suite 306W, Manchester, NH.

For more information about what happens when you report, how the university treats your information once a report is made to the Title IX Coordinator, your rights and reporting options at UNH (including anonymous reporting options) please visit [student reporting options](#). The uSafeUS app is also available for students to keep reporting options and resources easily accessible on their phones.

Help us improve our campus and community climate. If you have observed or experienced an incident of bias, discrimination or harassment, please report the incident by contacting the Civil Rights & Equity Office at UNH.civilrights@unh.edu or TEL # (603) 862-2930 voice/ (603) 862-1527 TTY / 7-1-1 Relay NH, or visit the CREO website. Anonymous reports may be submitted.

ABET/ETAC (Accreditation Board for Engineering and Technology/Engineering Technology Accreditation Commission)

I. General Criteria (2020)

Criterion 3. Student Outcomes

The program must have documented student outcomes with a documented and effective process for the periodic review and revision of these student outcomes. For purposes of this section, well-defined activities or problems are practical, narrow in scope, use conventional processes and materials in traditional ways, and require knowledge of standard operating processes. Broadly defined activities or problems are practical, broad in scope, relatively complex, and involve a variety of resources; use new processes, materials, or techniques in innovative ways; and may require extension of standard operating procedures.

B. For baccalaureate degree programs, these student outcomes must include, but are not limited to, the following:



- (1) an ability to apply knowledge, techniques, skills and modern tools of mathematics, science, engineering, and technology to solve broadly defined engineering problems appropriate to the discipline;
- (2) an ability to design systems, components, or processes meeting specified needs for broadly defined engineering problems appropriate to the discipline;
- (3) an ability to apply written, oral, and graphical communication in broadly defined technical and non-technical environments; and an ability to identify and use appropriate technical literature;
- (4) an ability to conduct standard tests, measurements, and experiments and to analyze and interpret the results to improve processes; and
- (5) an ability to function effectively as a member as well as a leader on technical teams.

II. PROGRAM CRITERIA

Discipline Specific Educational Outcomes

Electrical Engineering Technology

Lead Society: Institute of Electrical and Electronics Engineers

The UNH-M Electrical Technology program will prepare graduates with the skills necessary to enter careers in the design, application, installation, manufacturing, operation and/or maintenance of electrical/electronic(s) systems.

EET Outcomes

- a) The application of circuit analysis and design, computer programming, associated software, analog and digital electronics, and microcomputers, and engineering standards to the building, testing, operation, and maintenance of electrical/electronic(s) systems; and
- b) The application of natural sciences and mathematics at or above the level of algebra and trigonometry to the building, testing, operation, and maintenance of electrical/electronic systems.
- c) The ability to analyze, design, and implement one or more of the following: control systems, instrumentation systems, communications systems, computer systems, or power systems.
- d) The ability to apply project management techniques to electrical/electronic(s) systems; and
- e) The ability to utilize differential and integral calculus, as a minimum, to characterize the performance of electrical/electronic systems.